

AI MINOR COURSE — TECHNICAL REPORT

Optimizing Liquid-Liquid Extraction

Using Artificial Neural Networks (ANN)

A report submitted in partial fulfilment of the AI Minor Course

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Abstract

Liquid-liquid extraction (LLE) is one of the most commonly used separation techniques across industries including pharmaceuticals, food processing, and metallurgy. Optimising the process—determining the best combination of acid concentration, solvent ratio, and modifier composition—traditionally requires a large number of expensive and time-consuming experiments. This report demonstrates how Artificial Neural Networks (ANN), a branch of Artificial Intelligence, can be applied to accurately model and optimise LLE processes with far fewer experiments and significantly higher predictive accuracy. Using the reactive extraction of itaconic acid as a real-world case study, this report explains what ANN is, how it works, how it is applied to LLE, what results it achieves, and what its limitations are. The ANN model achieved a correlation coefficient (R^2) of 0.993, outperforming the traditional Response Surface Methodology ($R^2 = 0.970$), clearly demonstrating the power of AI in chemical engineering optimisation.

Keywords: *Liquid-liquid extraction, artificial neural network, ANN, itaconic acid, reactive extraction, process optimisation, machine learning, chemical engineering.*

1. Introduction

Chemical separation processes are at the heart of almost every manufacturing industry. Whether it is isolating a drug molecule in a pharmaceutical plant, recovering a metal ion from an ore, or purifying a food ingredient, engineers must find an efficient way to separate a desired compound from a mixture. Liquid-liquid extraction (LLE) is one of the simplest and most scalable ways to achieve this.

In LLE, a solute is transferred from one liquid phase—usually an aqueous phase containing water—into a second immiscible liquid phase called the organic phase. By choosing the right organic solvent and controlling conditions such as pH, temperature, and mixing ratio, engineers can selectively pull out the desired compound while leaving impurities behind. The challenge lies in finding the exact combination of process variables that gives the highest extraction efficiency with the least amount of solvent and energy.

Traditionally, this optimisation was done by changing one variable at a time and running hundreds of experiments. Later, statistical tools such as Response Surface Methodology (RSM) were introduced to reduce the number of experiments needed. However, RSM is based on polynomial equations and struggles to capture highly nonlinear relationships between variables—exactly the kind of relationships that arise in complex extraction systems.

This is where Artificial Intelligence, specifically Artificial Neural Networks (ANN), offers a powerful solution. ANNs are computational models inspired by the human brain. They are capable of learning complex, nonlinear patterns directly from experimental data without the need for any pre-assumed mathematical equation. Once trained, an ANN can accurately predict extraction efficiency for any combination of input variables—and can be coupled with an optimisation algorithm to find the best possible operating conditions automatically.

This report focuses on a specific, real published case study: the use of ANN to model and optimise the reactive extraction of itaconic acid using an organic phase containing tri-n-octylamine (TOA) as the extractant and dichloromethane (DCM) as the modifier. This example clearly illustrates the step-by-step application of ANN in LLE and demonstrates why AI outperforms traditional methods.

2. Background: What is Liquid-Liquid Extraction?

2.1 Principle of LLE

Liquid-liquid extraction is based on a simple idea: if a compound has a higher affinity for an organic solvent than for water, it will preferentially move into the organic phase when the two liquids are mixed and allowed to settle. The key measure of this preference is the distribution coefficient (D), defined as:

$$D = C_{\text{organic}} / C_{\text{aqueous}}$$

A higher D value means the solute strongly prefers the organic phase, resulting in high extraction efficiency. The percentage extraction (% E) is related to D by:

$$\% E = 100 \times D / (D + V_{aq}/V_{org})$$

where V_{aq} and V_{org} are the volumes of the aqueous and organic phases respectively. In practice, engineers want to maximise % E while keeping solvent usage and cost as low as possible.

2.2 Key Process Variables

The main variables that control extraction efficiency in LLE are:

- Solute concentration in the feed (aqueous phase)
- Extractant concentration — the organic compound that chemically binds to the solute
- Modifier concentration — an additional organic compound added to improve solubility
- pH of the aqueous phase
- Temperature and mixing intensity
- Phase ratio (volume of organic to volume of aqueous)

Because these variables interact in complex, nonlinear ways, traditional mathematical models are often unable to accurately predict how changing one variable will affect extraction when others are also changing simultaneously.

3. What is an Artificial Neural Network?

3.1 Inspiration from the Human Brain

The human brain contains approximately 86 billion neurons — nerve cells that are connected to each other and transmit signals. When we learn something, the connections (synapses) between neurons are strengthened or weakened based on experience. An Artificial Neural Network mimics this learning process using mathematical operations inside a computer.

An ANN consists of layers of simple processing units called artificial neurons (or nodes). Each neuron receives numerical inputs, multiplies them by learnable weights, adds a bias term, and passes the result through a mathematical function called an activation function. The output of one layer becomes the input to the next.

3.2 Structure of a Feedforward ANN

The most common type of ANN used in engineering is the Feedforward Neural Network (also called a Multilayer Perceptron). It has three types of layers:

- Input Layer — receives the raw experimental data (e.g., acid concentration, extractant %, modifier %)
- Hidden Layer(s) — performs all internal computations; this is where the ANN 'learns' the nonlinear patterns
- Output Layer — produces the final prediction (e.g., extraction efficiency %)

The diagram below shows the structure of the ANN used in the case study in this report:

Layer	Neurons	Role in LLE Case Study
Input Layer	3 neurons	Acid concentration, Extractant (TOA) %, Modifier (DCM) %
Hidden Layer	10 neurons	Learns nonlinear relationships between inputs and output
Output Layer	1 neuron	Predicted extraction efficiency (% Y)

3.3 How Training Works

Training is the process by which an ANN learns from data. It works as follows:

- Step 1 — Collect experimental data: run a set of lab experiments with known input-output combinations
- Step 2 — Forward pass: feed the inputs into the ANN and let it predict an output
- Step 3 — Calculate error: compare the ANN's prediction against the actual experimental result

- Step 4 — Backpropagation: mathematically calculate how much each weight contributed to the error
- Step 5 — Update weights: adjust all weights slightly to reduce the error
- Step 6 — Repeat steps 2–5 over many iterations (epochs) until the error is minimised

The most widely used training algorithm for LLE applications is the Levenberg-Marquardt (LM) backpropagation algorithm, which combines the speed of gradient descent with the stability of the Gauss-Newton method. Once trained, the ANN can predict the output for any new set of inputs instantly—without running another experiment.

3.4 Why ANN is Better than Traditional Methods for LLE

Feature	Traditional RSM	Artificial Neural Network (ANN)
Model type	Fixed polynomial equation	Flexible nonlinear mapping
Handles complex interactions?	Partially (only up to 2nd order)	Yes, any level of complexity
Number of experiments needed	More (full factorial design)	Fewer (can generalise)
Prediction accuracy (R^2)	~0.97	~0.99
Can be coupled with optimiser?	Yes (limited)	Yes (very effectively with GA)
Requires prior knowledge of model?	Yes	No — learns from data

4. Case Study: ANN for Reactive Extraction of Itaconic Acid

4.1 What is Itaconic Acid and Why Extract It?

Itaconic acid (IA) is an organic dicarboxylic acid produced by fungal fermentation. It is used as a monomer in the production of synthetic fibres, adhesives, detergents, and superabsorbent polymers. The global market for itaconic acid has been growing rapidly due to its bio-based, renewable nature. However, recovering itaconic acid from a dilute fermentation broth (aqueous solution) is a major challenge that significantly adds to production costs.

Reactive liquid-liquid extraction is one of the most efficient methods for this recovery. In reactive extraction, an extractant (TOA — tri-n-octylamine) reacts chemically with the itaconic acid molecules in the aqueous phase, forming an extractant-acid complex that then migrates into the organic phase. A modifier (DCM — dichloromethane) is added to the organic phase to improve the solubility of the extracted complex and avoid third-phase formation.

The three key variables that control the extraction percentage (%Y) in this system are:

- Itaconic acid concentration in the feed — ranging from 0.036 to 0.108 mol/L
- Extractant (TOA) composition in organic phase — ranging from 5% to 30% v/v
- Modifier (DCM) composition in organic phase — ranging from 30% to 90% v/v

4.2 Experimental Data Collection

The researchers designed a set of experiments using a statistical approach called the Box-Behnken Design (BBD), which allows the relationships among three variables to be studied efficiently with a minimum number of experiments. A total of 15 experiments were conducted, varying the three input variables at three levels each (low, medium, high). The output measured in each experiment was the percentage extraction (%Y) of itaconic acid into the organic phase.

This experimental data — 15 rows of [acid conc., TOA %, DCM %] → [%Y] — became the training dataset for both the ANN model and the RSM model. The dataset was split into a training set (used to build the model) and a validation set (used to check accuracy on unseen data).

4.3 Building and Training the ANN Model

The ANN was constructed with the following architecture, selected after systematic testing of different configurations to find the minimum error:

Parameter	Value / Choice	Reason
Input neurons	3	Three input variables: acid conc., TOA %, DCM %
Hidden layers	1	Sufficient for this complexity level

Parameter	Value / Choice	Reason
Neurons in hidden layer	10	Minimised MSE during optimisation testing
Output neurons	1	Single output: extraction efficiency % Y
Activation function	Sigmoid (tansig)	Standard for nonlinear regression tasks
Training algorithm	Levenberg-Marquardt	Fast convergence for small datasets
Performance metric	Mean Squared Error (MSE)	Measures average prediction error

The network was implemented in MATLAB using the Neural Network Toolbox. The training process adjusted all connection weights iteratively using backpropagation until the MSE on the training set reached its minimum. The final trained model was then tested on the validation set to confirm that it had not simply memorised the training data (a problem called overfitting) but had genuinely learned the underlying relationship.

4.4 Results: ANN vs. RSM Performance

After training, the ANN and RSM models were compared on their ability to predict extraction efficiency. The key performance metric used was R^2 (coefficient of determination), which measures how close the model's predictions are to the actual experimental values. An R^2 of 1.0 means perfect prediction; higher is better.

Metric	RSM Model	ANN Model	Winner
R^2 (correlation coefficient)	0.970	0.993	ANN
Training accuracy	Good	Excellent	ANN
Validation accuracy	Good	Excellent	ANN
Handles nonlinearity	Partially	Fully	ANN
Model interpretability	High (equation given)	Low (black box)	RSM
Computation time	Very fast	Moderate	RSM

The ANN model's R^2 of 0.993 means that 99.3% of the variation in extraction efficiency across all 15 experiments was correctly explained by the model. This is a remarkably high accuracy for a complex chemical system and clearly demonstrates the superiority of ANN over RSM for this application.

4.5 Optimum Conditions Found by ANN

Once the ANN was trained and validated, it was used as a surrogate model—meaning it replaced the expensive lab experiments in an optimisation search. The optimal design factors identified for maximum itaconic acid extraction were:

Variable	Optimum Value	Range Tested
Itaconic acid concentration	0.072 mol/L	0.036 – 0.108 mol/L
Extractant (TOA) composition	16.075 % v/v	5 – 30 % v/v
Modifier (DCM) composition	62.15 % v/v	30 – 90 % v/v
Predicted extraction % Y (ANN)	100.69 %	—
Experimental extraction % Y	98.86 %	—

The close match between the ANN-predicted value (100.69%) and the actual experimental result (98.86%) confirms the model's practical reliability. These optimal conditions mean that by using an intermediate concentration of the extractant (not too much, not too little) and a moderately high modifier concentration, engineers can extract nearly all of the itaconic acid from the fermentation broth in a single extraction step—a highly significant result for industrial scale-up.

An important physical insight also emerged from the ANN simulation results: using too high a concentration of TOA actually reduced extraction efficiency, because excess TOA increased the viscosity of the organic phase, slowing down mass transfer. This kind of insight—hard to obtain from simple trial-and-error—demonstrates the additional value that ANN modelling brings beyond just prediction.

5. Advantages and Limitations of ANN in LLE

5.1 Advantages

- Universal approximator: ANN can in principle model any nonlinear function, making it applicable to virtually any LLE system regardless of the chemistry involved.
- No prior knowledge required: unlike mechanistic models (e.g., NRTL or UNIQUAC thermodynamic models), ANN does not require the engineer to know the physical mechanism behind the extraction. It learns purely from data.
- Higher accuracy: as shown in the case study, ANN consistently achieves higher R^2 values than RSM for the same dataset, particularly when variable interactions are complex.
- Fast prediction after training: once the ANN is trained, it can predict extraction efficiency for any new set of conditions in milliseconds—far faster than running a new experiment.
- Easily coupled with optimisers: ANN models can be combined with genetic algorithms (GA), particle swarm optimisation (PSO), or Bayesian optimisation to automatically search for the global optimum across a wide variable space.
- Adaptable to multi-output problems: the output layer of an ANN can be extended to simultaneously predict multiple outputs (e.g., extraction efficiency AND selectivity AND solvent loss), which is very useful for multi-objective optimisation.

5.2 Limitations

- Requires sufficient data: an ANN needs a reasonable amount of experimental data to train reliably. With very few data points (fewer than 10–15), the model may overfit—memorising the training data rather than learning the true relationship.
- Black-box nature: unlike RSM which gives a clear algebraic equation, a trained ANN is difficult to interpret. Engineers cannot easily see why it predicts a certain value, which can reduce confidence in the model, especially for regulatory applications.
- Sensitive to input scaling: ANN training is sensitive to the range of input values. All inputs must be normalised (scaled to 0–1 or –1 to +1) before training, otherwise neurons with large-valued inputs will dominate the learning process.
- Risk of local minima: the backpropagation training algorithm can sometimes converge to a local minimum rather than the global minimum of the error surface, meaning the final model may not be the best possible one. This can be mitigated by running training multiple times with different random initialisations.
- Limited extrapolation: ANN models are only reliable within the range of variables covered by the training data. Predictions made far outside the trained range (extrapolation) can be wildly inaccurate.
- Architecture selection is non-trivial: choosing the optimal number of hidden layers and neurons requires systematic testing or prior experience, adding complexity to the model development process.

6. Future Directions

The application of AI to LLE optimisation is still a rapidly evolving field. Several exciting directions are emerging that will further enhance the capabilities demonstrated in this report.

6.1 Hybrid Physics-Informed ANN

One of the most promising developments is the creation of hybrid models that combine traditional thermodynamic equations (such as NRTL or UNIQUAC activity coefficient models) with ANN. In these models, the physics-based equations provide the ANN with prior knowledge of the general chemical behaviour, reducing the amount of experimental data needed for training and improving generalisation beyond the training range.

6.2 Transfer Learning for New Systems

Transfer learning—a technique where a model trained on one chemical system is fine-tuned for a related but different system—could dramatically reduce the data requirements for LLE optimisation. An ANN pre-trained on copper extraction data, for example, could be quickly adapted for neodymium extraction with only a small number of new experiments.

6.3 Digital Twins of Extraction Columns

A digital twin is a real-time virtual replica of a physical extraction column, powered by a continuously updated ANN model. As sensor data from the real column (flow rates, temperatures, concentrations) streams in, the ANN updates its predictions and automatically suggests adjustments to process variables to maintain optimal performance. This approach is already being explored in major chemical companies.

6.4 Autonomous Self-Driving Laboratories

The most futuristic direction is the fully automated laboratory where a robotic system performs experiments, feeds the data directly to an ANN, and uses an optimisation algorithm to decide what experiment to run next—all without human intervention. This closed-loop approach can find optimal extraction conditions in a fraction of the time and cost of traditional methods.

7. Conclusion

This report has provided a clear, step-by-step explanation of how Artificial Neural Networks can be applied to optimise liquid-liquid extraction processes. Starting from the basic principles of LLE and ANN, the report walked through a real published case study—the reactive extraction of itaconic acid using TOA and DCM—to demonstrate the entire workflow: data collection, ANN construction, training, validation, and optimisation.

The results of the case study unambiguously show that ANN outperforms the traditional Response Surface Methodology, achieving an R^2 of 0.993 compared to 0.970 for RSM. The optimal process conditions identified by ANN (0.072 mol/L acid, 16.075% TOA, 62.15% DCM) matched experimental results to within 1.85%, confirming the model's practical reliability for industrial application.

While ANN has clear advantages—high accuracy, flexibility, no need for prior mechanistic knowledge, and easy integration with optimisation algorithms—its limitations must also be acknowledged. The black-box nature of ANN reduces interpretability, and care must be taken to ensure sufficient training data, proper input normalisation, and validation against unseen data.

Looking ahead, the integration of ANN with physics-based models, transfer learning, digital twins, and autonomous laboratories promises to make AI-driven LLE optimisation even more powerful and accessible. For students and engineers entering the field today, understanding how to apply ANN to chemical process optimisation is an increasingly essential skill—one that sits at the intersection of classical chemical engineering and the emerging discipline of AI.

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